An Exact Histogram for a Quadratic Staircase

Paper 1. The increments of $\lfloor 2j^2/n \rfloor$, and the local maxima of $2j^2 \bmod n$

Abstract

For odd $n \geq 3$ let

$$W_j = \left\lfloor \frac{2(j+1)^2}{n} \right\rfloor - \left\lfloor \frac{2j^2}{n} \right\rfloor, \qquad j = 0, 1, \ldots, n-1.$$

Main result (Theorem 2). Every W_j lies in $\{0,1,2,3,4\}$, and with $A = \lfloor (n+7)/8 \rfloor$ the five values occur exactly

$$\text{card}\{W{=}0\} = \text{card}\{W{=}4\} = A, \qquad \text{card}\{W{=}2\} = 2A{-}1, \qquad \text{card}\{W{=}1\} = \text{card}\{W{=}3\} = \tfrac{n+1}{2} - 2A.$$

There is no error term. The proof is a tiling, and the residues $2j^2 \bmod n$ never enter it. A random model of the residues gives the main term $n/8$; it stays within 1 of the true count but does not equal it.

The same tiling also says *where* the values 0 and 2 fall: at the floors of the rising and falling level crossings of the parabola $x(n-2x)/n$ (Theorem 2b). Symmetric pairs compress to one symbol each, with a census again fixed by one integer (Theorems 3 and 4).

Second result (Theorem 5). The sequence $2j^2 \bmod n$ has exactly $2A$ interior local maxima for every odd n except $n = 3, 5, 7, 9, 49$. The proof is uniform for $n \geq 51$; the smaller cases are checked one by one.

Nothing in this paper concerns prime numbers.

How to read the claims in this paper. Statements set as Theorems, Propositions and Corollaries are proved, and the proofs are given. Everything else falls into two kinds, and we try to keep them apart. A *measurement* is a computation over a stated finite range; it is labelled with that range, and it supports a claim about that range only. A *reading* is our own judgement about what a measurement or a proof appears to mean, and we mark it as ours rather than stating it as established. In this paper the proportion is heavily weighted to the first kind: apart from the sample tables, the one item that is a measurement rather than a proof is the control comparison of §3.2, and it is labelled there.

Keywords: floor function, quadratic residues, exact multiplicities, three-distance theorem, equidistribution.

MSC 2020: 11B57, 11A07, 11K31.

1. The object

Throughout, $n \geq 3$ is odd and j runs over $0, 1, \ldots, n-1$. Set

$$F(x) = \left\lfloor \frac{2x^2}{n} \right\rfloor, \qquad W_j = F(j+1) - F(j), \qquad r_j = 2j^2 \bmod n.$$

F is a staircase under the parabola $2x^2/n$; W_j is its increment, and r_j is the residue left over. The two are linked by $2j^2 = nF(j) + r_j$, but — and this is the point of §3 — the histogram of W can be determined without ever computing an r_j.

Two elementary remarks fix the scale. The total rise is

$$\sum_{j=0}^{n-1} W_j = F(n) - F(0) = 2n,$$

so the mean increment is exactly 2; and W_j counts the multiples of n in the interval

$$I_j = (2j^2,\ 2(j+1)^2], \qquad |I_j| = 4j + 2. \qquad (1.1)$$

The intervals $I_0, I_1, \ldots$ tile $(0, \infty)$. That is the only structural fact the histogram needs.

---

2. Theorem 1: the range

Theorem 1. For every odd n and every $0 \le j \le n-1$, $W_j \in \{0, 1, 2, 3, 4\}$.

Proof. The two arguments of the floor differ by

$$\frac{2(j+1)^2}{n} - \frac{2j^2}{n} = \frac{4j+2}{n},$$

which is positive and, for $j \le n-1$, strictly less than 4. A difference of floors of two reals differing by less than 4 lies in $\{0, 1, 2, 3, 4\}$. ■

Verification. Zero violations over every odd $n \le 20{,}001$.

The restriction $j \le n-1$ is real: for j in the range $[rn, (r+1)n)$ the same computation gives $W_j \in \{4r, \ldots, 4r+4\}$, so the bound moves up in steps of four. Everything below concerns $0 \le j \le n-1$, which we call the **first cycle**.

Two further symmetries will be used. Both are immediate from $\lfloor 2(n-x)^2/n \rfloor = \lfloor 2x^2/n \rfloor + 2n - 4x$, which gives $W_{n-1-j} = 4 - W_j$:

$$W_j + W_{n-1-j} = 4, \qquad W_{(n-1)/2} = 2. \qquad (2.1)$$

---

3. Theorem 2: the exact multiplicities

Theorem 2. Let $n \ge 3$ be odd and $A = \lfloor (n+7)/8 \rfloor$. Over the first cycle, $\mathrm{card}\{W{=}0\} = \mathrm{card}\{W{=}4\} = A$, $\mathrm{card}\{W{=}2\} = 2A - 1$, $\mathrm{card}\{W{=}1\} = \mathrm{card}\{W{=}3\} = \dfrac{n+1}{2} - 2A$, and no other value occurs.

n	$(N_0, N_1, N_2, N_3, N_4)$	A
11	$(2, 2, 3, 2, 2)$	2
101	$(13, 25, 25, 25, 13)$	13
1,009	$(127, 251, 253, 251, 127)$	127
19,997	$(2500, 4999, 4999, 4999, 2500)$	2500

Verification. Zero exceptions for every odd $n \le 200{,}001$.

Proof. By (1.1), W_j counts the multiples of n in I_j, an interval of length exactly $4j+2$. Hence:

- if $4j+2 < n$, the interval is shorter than n and holds **at most one** multiple, so $W_j \in \{0, 1\}$;
- if $4j+2 \ge n$, it is at least as long and holds **at least one**, so $W_j \ge 1$.

The count of zeros. Consequently $W_j = 0$ forces $j < m := \lfloor (n+1)/4 \rfloor$. On that range the intervals $I_0, \ldots, I_{m-1}$ tile $(0, 2m^2]$ exactly and each holds at most one multiple, so the number of *occupied* intervals equals the number of multiples in the range, namely $\lfloor 2m^2/n \rfloor$. Therefore

$$N_0 = m - \left\lfloor \frac{2m^2}{n} \right\rfloor. \qquad (3.1)$$

Write $n = 8t + r$ with $r \in \{1, 3, 5, 7\}$; then $m = 2t + \varepsilon$ with $\varepsilon = 0, 1, 1, 2$ respectively. We claim the quotient in (3.1) is $q = t + \varepsilon - 1$:

r	m	q	$2m^2 - nq$	check
1	$2t$	$t-1$	$7t+1$	$< 8t+1$ ✓
3	$2t+1$	t	$5t+2$	$< 8t+3$ ✓
5	$2t+1$	t	$3t+2$	$< 8t+5$ ✓
7	$2t+2$	$t+1$	$t+1$	$< 8t+7$ ✓

Each remainder is a positive linear function of t strictly below n, so the quotient is as claimed and

$$N_0 = (2t+\varepsilon) - (t+\varepsilon-1) = t+1 = \left\lfloor \frac{n+7}{8} \right\rfloor = A.$$

The remaining four. Put $H = (n-1)/2$. For $0 \le j < H$ the interval I_j has length $4j+2 < 2n$, so $W_j \in \{0, 1, 2\}$. These H intervals tile $(0, 2H^2]$, which contains

$$\left\lfloor \frac{2H^2}{n} \right\rfloor = H - 1$$

multiples of n. Writing $N_i^- = \mathrm{card}\{0 \le j < H : W_j = i\}$, we have $N_0^- = A$ together with

$$N_0^- + N_1^- + N_2^- = H, \qquad N_1^- + 2N_2^- = H - 1,$$

and subtracting gives $N_2^- = A - 1$. The central index $j = H$ has $W_H = 2$ by (2.1). Finally the symmetry $W_j + W_{n-1-j} = 4$ pairs $W{=}0$ with $W{=}4$ and $W{=}1$ with $W{=}3$, while $W{=}2$ pairs with itself. Therefore

$$N_4 = N_0 = A, \qquad N_2 = 2(A-1) + 1 = 2A - 1,$$

and the remaining $n-(N_0+N_2+N_4)$ indices split equally:

$$N_1 = N_3 = \frac{n+1}{2} - 2A. \qquad \blacksquare$$

3.1 Why the count has no error term

A probabilistic model, assuming $2j^2 \bmod n$ equidistributed, estimates N_0 by

$$\sum_{j<(n-2)/4} \frac{n-4j-2}{n} = \frac{n^2-1}{8n},$$

the sum being exactly that for every odd n, since it telescopes to $m - 2m^2/n$ with $m = \lfloor (n+1)/4 \rfloor$. The true value is $N_0 = A = \lceil n/8 \rceil$, so the model returns the main term $n/8$ and not the value itself: the difference $A - (n^2-1)/(8n)$ lies in $[0,1)$ for every odd n and reaches 0.889 at $n=9$, where the model gives $10/9$ against a true count of 2.

The exactness of Theorem 2 is a property of the tiling and not of any averaging. The intervals I_j partition the range, so the number of occupied ones is a difference of two floors taken at the two ends, and nothing is estimated anywhere in the argument. The residues $2j^2 \bmod n$ never enter it. Theorem 5 is the statement about those residues, and it is a good deal harder.

The same absence, named, and in a more general form. Posing the question on MathOverflow produced a framing that identifies the mechanism and extends it to $\lfloor kj^2/n \rfloor$ for any k. Group the j by the value of the linear part, $J_q = \{j : \lfloor k(2j+1)/n \rfloor = q\}$, an interval with endpoints

$$a_q = \left\lceil \frac{qn-k}{2k} \right\rceil, \qquad b_q = \left\lceil \frac{(q+1)n-k}{2k} \right\rceil - 1,$$

and let $L_q = |J_q|$. Define

$$C_q = \left\lfloor \frac{k(b_q+1)^2}{n} \right\rfloor - \left\lfloor \frac{ka_q^2}{n} \right\rfloor - qL_q,$$

which by telescoping $\sum_{j\in J_q}(W_j - q)$ is exactly the number of **carries** on J_q — the j at which the increment is $q+1$ rather than q. The multiplicities are then

$$N_0 = L_0 - C_0, \qquad N_i = L_i - C_i + C_{i-1} \quad (1 \le i < 2k), \qquad N_{2k} = C_{2k-1}.$$

These three follow in one step, and we record the step rather than leave the identity as a measurement. On J_q the real increment $k(2j+1)/n$ lies in $[q, q+1)$, so $W_j \in \{q, q+1\}$ there, and $C_q = \sum_{j\in J_q}(W_j - q)$ counts exactly the $j \in J_q$ at which the larger value is taken. An index with $W_j = i$ therefore lies either in J_i and is not a carry, or in J_{i-1} and is one, which gives the middle formula; the two ends are the cases where only one of the two alternatives exists.

Verification. Zero mismatches over 17,991 pairs (k,n) with $2 \le k \le 10$ and odd $n < 4000$, over $k = 50, 100, 200, 400$ with odd $n \le 401$, and over 3,987 pairs with **even** n ($2 \le k \le 10$, $n \le 888$), which Theorem 2 excludes. For $k=2$ it reproduces the four multiplicities of Theorem 2 exactly.

So the absence of an error term is a carry-count phenomenon rather than an equidistribution one, which is the same reading as the tiling above, stated in a form that survives the generalisation in k. *We have not found this recorded anywhere; the question asking for a reference is still open, and we give the identity with its one-step proof above rather than as a known result.* The framing is due to a MathOverflow answer [4]; the verification and the extension to even n are ours.

3.2 Theorem 2b: where the increments are, not only how many

Theorem 2 counts the five values. They can also be *located*, in closed form, and the argument is the tiling of §1 read geometrically rather than arithmetically.

Set

$$D_j = j - \left\lfloor \frac{2j^2}{n} \right\rfloor, \qquad \text{so that} \qquad D_{j+1} - D_j = 1 - W_j.$$

Because j is an integer, $D_j = \lceil f(j) \rceil$ with

$$f(x) = x - \frac{2x^2}{n} = \frac{x(n-2x)}{n},$$

a concave parabola with its peak at $x = n/4$. So D is the ceiling quantisation of a parabola, $W_j = 0$ means D rises, $W_j = 1$ that it is flat, and $W_j = 2$ that it falls. The quantity quantised is D_j, not the staircase $\lfloor 2j^2/n \rfloor$ itself: the staircase is only a non-decreasing step function under the convex parabola $2x^2/n$, and it is not convex.

> **Theorem 2b.** Let $H = (n-1)/2$ and $A = \lfloor (n+7)/8 \rfloor$ as before, and for $1 \le m \le A$ put $\alpha_m = \left\lfloor \dfrac{n - \sqrt{n^2 - 8n(m-1)}}{4} \right\rfloor$, $\beta_m = \left\lceil \dfrac{n + \sqrt{n^2 - 8n(m-1)}}{4} \right\rceil - 1$. Then, for $0 \le j < H$, $W_j = 0 \iff j = \alpha_m$ for some $1 \le m \le A$, $W_j = 2 \iff j = \beta_m$ for some $2 \le m \le A$, and $W_j = 1$ at every other index of that range. Moreover $\max_j D_j = A$, and the two families are separated by the peak: every $\alpha_m < n/4$ and every $\beta_m + 1 > n/4$.

Proof. $f(j+1) - f(j) = (n - 4j - 2)/n$ has absolute value less than 1 throughout $0 \le j < H$, so the ceiling can move by at most one step and D is unimodal: it rises while $j < n/4$ and falls after, giving the separation and leaving only $W \in \{0, 1, 2\}$ on this range. A rise into level m happens at the last integer before f crosses the value $m - 1$ upward, and a fall out of it at the last integer before f crosses $m - 1$ downward; solving $f(x) = m - 1$ gives the two roots

$$x_m^{\pm} = \frac{n \pm \sqrt{n^2 - 8n(m-1)}}{4},$$

whence the stated floor and ceiling. The peak value of f on the integers is $(n^2 - 1)/(8n)$, since n is odd and the nearest integer to $n/4$ is at distance $1/4$; its ceiling is A. ■

Verification. Zero mismatches for all 10,000 odd n up to 20,001, and the count $|\{\alpha_m\}| = A$ recovers $N_0 = A$ of Theorem 2 for every odd $n \le 20{,}001$. The separation at the peak has zero violations for odd $n < 4{,}000$.

A mirror that halves the data. The two roots satisfy $x_m^- + x_m^+ = n/2 = H + \frac{1}{2}$, and after the floor and the ceiling this becomes

$$\beta_m = H - \alpha_m - \varepsilon_m, \qquad \text{where} \qquad \varepsilon_m = \begin{cases} 0, & \{x_m^-\} < \frac{1}{2}, \\ 1, & \{x_m^-\} \ge \frac{1}{2}, \end{cases}$$

which is immediate: writing $x_m^- = \alpha_m + t$ with $0 \le t < 1$, the root sum gives $x_m^+ = H + \frac{1}{2} - \alpha_m - t$, so $\beta_m = \lceil x_m^+ \rceil - 1 = H - \alpha_m + \lceil \frac{1}{2} - t \rceil - 1$, and $\lceil \frac{1}{2} - t \rceil$ is 1 for $t < \frac{1}{2}$ and 0 otherwise. Verified in that form with zero mismatches over 70,687 pairs (n, m) with $n < 1{,}500$. So the descending half

needs no separate list: the ascending positions α_m and one bit per level determine everything, and the palindrome $W_j + W_{n-1-j} = 4$ of (2.1) then determines the whole word.

What this does not give. The crossing set $\mathcal{A}_n = \{\alpha_m\}$ is a compression of the word, not an arithmetic invariant of it, and nothing in Theorem 2b relates $\mathcal{A}_n$ mod q to n at a fixed q. A χ^2 comparison of $\mathcal{A}_n$ mod q at $q = 7, \dots, 23$, over odd $n \in (10^3, 2 \cdot 10^4)$, against a control that keeps the crossing density, showed no sign of extra structure. The control was not recorded in a form that can be rerun and no generator was kept, so the comparison supports nothing; it is mentioned only because the closed form invites the opposite guess.

---

4. Theorems 3 and 4: pairs

Index the symmetric pairs $\{j, n-1-j\}$ by $R = 1, 3, \dots, n-2$, the pair at R being the one whose upper index is $j = (n+R)/2$, so that $W_{(n+R)/2} = 2 + d_n(R)$ and its partner sits at $n-1-j$.

Theorem 3. With $d_n(R) = \left\lfloor \frac{1}{2} + \frac{(R+2)^2}{2n} \right\rfloor - \left\lfloor \frac{1}{2} + \frac{R^2}{2n} \right\rfloor$, one has $d_n(R) \in \{0, 1, 2\}$, and the corresponding pair of increments is $(W_L, W_R) = (2 - d,\ 2 + d)$.

Proof. The two arguments of the floor differ by

$$\frac{(R+2)^2 - R^2}{2n} = \frac{2(R+1)}{n},$$

which for $1 \le R \le n-2$ lies strictly between 0 and 2; hence the difference of floors is 0, 1 or 2. Substituting the pair parametrisation into the definition of W gives $W_R = 2 + d_n(R)$, and (2.1) then gives $W_L = 2 - d_n(R)$. ■

Verification. Zero failures over every odd $n < 600$ and every odd R with $1 \le R \le n-2$.

Theorem 4. Let $P_d = \mathrm{card}\{R : d_n(R) = d\}$, written P rather than N to keep it apart from the multiplicities of Theorem 2. Then $P_2 = P_0 + 1$.

Proof. There are $(n-1)/2$ odd values $R = 1, 3, \dots, n-2$, and the definition of $d_n(R)$ telescopes along them:

$$\sum_R d_n(R) = \left\lfloor \tfrac{1}{2} + \frac{n^2}{2n} \right\rfloor - \left\lfloor \tfrac{1}{2} + \frac{1}{2n} \right\rfloor = \frac{n+1}{2}.$$

Hence

$$P_0 + P_1 + P_2 = \frac{n-1}{2}, \qquad P_1 + 2P_2 = \frac{n+1}{2},$$

and subtracting gives $P_2 - P_0 = 1$. ■

Verification. Zero failures over all odd $n < 2000$.

n	P_0	P_1	P_2
11	1	2	2
13	1	3	2
31	3	8	4
101	12	25	13
499	62	124	63

The whole census, from Theorem 2. The partner of the upper index $(n+R)/2$ is $i = (n-2-R)/2$, so $W_i = 2 - d_n(R)$ by (2.1). Hence $d_n(R) = 2$ exactly when $W_i = 0$, and $d_n(R) = 0$ exactly when $W_i = 2$ with $i < (n-1)/2$. Below the middle the real increment $2(2j+1)/n$ is less than 2, so $W_j \le 2$ there and, by (2.1), the value 0 never occurs above the middle: all A zeros of Theorem 2 are lower indices. The value 2 occurs $2A-1$ times, once at the middle and otherwise in mirror pairs. So

$$P_2 = A, \qquad P_0 = A - 1, \qquad P_1 = \tfrac{n+1}{2} - 2A,$$

which contains Theorem 4 and fixes each count, not only their difference; the rows of the table are these values. As in Theorem 2, the whole census is governed by one integer.

---

5. Theorem 5: the number of local maxima of $2j^2 \bmod n$

Theorem 2 counts the values of W. The companion question concerns their *arrangement*, and its sharpest form is the number of local maxima of the residue sequence itself.

> **Theorem 5.** Let $n \ge 3$ be odd and $A = \lfloor (n+7)/8 \rfloor$. Then $r_j = 2j^2 \bmod n$, $j = 0, \dots, n-1$, has exactly $2A$ interior local maxima, except at the five values $n = 3, 5, 7, 9, 49$.

The maximum is meant strictly: j is counted when $r_j > r_{j-1}$ and $r_j > r_{j+1}$. The proof below reads a maximum through the weaker pair of conditions $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$, and the two readings agree from $n = 11$ on.

> **Lemma (the two readings of a maximum).** For odd $n \ge 11$ and $1 \le j \le n-2$, the conditions $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$ hold together if and only if j is a strict local maximum.

Proof. The two readings can differ only where $r_j = r_{j-1}$. That forces $2j^2 \equiv 2(j-1)^2$, i.e. $n \mid 2(2j-1)$, and n is odd, so $n \mid 2j-1$; for $1 \le j \le n-1$ the only solution is $j = H+1$ with $H = (n-1)/2$. Doubling identifies the two residues there: $2r_H \equiv (n-1)^2 \equiv 1$ and $2 \cdot \frac{n+1}{2} = n+1 \equiv 1$, so $r_H = r_{H+1} = (n+1)/2$; and $2r_{H-1} \equiv (n-3)^2 \equiv 9$ with $2 \cdot \frac{n+9}{2} = n+9 \equiv 9$, so $r_{H-1} = (n+9)/2$ as soon as $(n+9)/2 < n$, that is $n > 9$. Since $r_{H+2} = r_{n-H-2} = r_{H-1}$, we get $r_{H+2} > r_{H+1}$, so $\varepsilon_{H+1} = 0$ and the weaker reading does not fire at $j = H+1$ either. ■

At $n = 5, 7, 9$ the quantity $(n+9)/2$ is not below n, the last step fails, and the two readings do differ; all three are among the five exceptions in any case.

Range of the statement. The argument below is uniform for $n \ge 51$. Two things need it: the interval structure of Step 4 is in its generic configuration once $n \ge 13$, and the Lemma of §5.5 needs $e_x^2 < n$, which holds exactly when $n > 49$. The twenty-four odd values $3 \le n \le 49$ are settled instead by direct computation, and five of them fail — $n = 3, 5, 7, 9, 49$ — so the statement above, exception

list included, is proved rather than observed. The count was checked against the direct definition for all odd $n \le 200{,}001$ as well.

Throughout put

$$\varepsilon_j = [\, r_{j+1} < r_j \,], \qquad q_j = \left\lfloor \frac{4j+2}{n} \right\rfloor, \qquad H = \frac{n-1}{2},$$

$$\Phi(x) = \left\lfloor \frac{2x^2}{n} \right\rfloor + \left\lfloor \frac{2(x-1)^2}{n} \right\rfloor,$$

so that $\varepsilon_j = W_j - q_j$, and j is a local maximum exactly when $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$.

5.1 Step 1: the palindrome

Since $2(n-j)^2 \equiv 2j^2$, we have $r_j = r_{n-j}$ for $1 \le j \le n-1$. Maxima therefore occur in mirror pairs $j \leftrightarrow n-j$, with no fixed point because n is odd. The pairing stays inside the interior indices because $j = 1$ carries no maximum once $n \ge 9$, where $r_1 = 2$ and $r_2 = 8$; at $n = 7$ it does carry one, and its mirror $n-1$ is not an interior index, which is one of the reasons 7 is an exception. Also $r_{H+1} = r_{n-H-1} = r_H$, so

$$\varepsilon_0 = 0 \quad (\text{since } r_1 = 2 > 0 = r_0), \qquad \varepsilon_H = 0,$$

and no maximum straddles the middle. Hence the total is twice the number of maxima in $1 \le j \le H$.

5.2 Step 2: maxima are exactly the mixed pairs

In any 0–1 word the numbers of ascents and descents differ by the boundary values; applied to $\varepsilon_0, \ldots, \varepsilon_H$ with both ends 0, the two are equal. Therefore

$$\operatorname{card}\{j \le H : \varepsilon_{j-1} = 0,\ \varepsilon_j = 1\} = \tfrac{1}{2}\operatorname{card}\{j \le H : \varepsilon_{j-1} \ne \varepsilon_j\},$$

and combining with Step 1:

$$\boxed{\operatorname{card}\{\text{local maxima}\} = \operatorname{card}\{1 \le j \le H : \varepsilon_{j-1} \ne \varepsilon_j\}.}$$

5.3 Step 3: a mixed pair is one condition on one interval

Since $\varepsilon_{j-1} + \varepsilon_j = (W_{j-1} + W_j) - (q_{j-1} + q_j)$, the pair at j is mixed if and only if

$$P_j := \big(2(j-1)^2,\ 2(j+1)^2\big] \quad \text{contains exactly } q_{j-1} + q_j + 1 \text{ multiples of } n.$$

P_j has length exactly $8j$. No residue occurs in this condition.

5.4 Step 4: the pair intervals tile, by parity

Consecutive P_j overlap, but each parity subfamily tiles exactly:

$$P_{2s+1} = \big(8s^2,\ 8(s+1)^2\big], \qquad P_{2s} = \big(2(2s-1)^2,\ 2(2s+1)^2\big],$$

of lengths $8(2s+1)$ and $16s$. Since $8j < 4n$ on $j \le H$, each P_j carries L_j or $L_j + 1$ multiples with $L_j = \lfloor 8j/n \rfloor \in \{0, 1, 2, 3\}$, and over any block of consecutive j of one parity the total telescopes. Set $\mathrm{off}_j = (q_{j-1} + q_j) - L_j$; then j is mixed iff P_j carries the **larger** count when $\mathrm{off}_j = 0$, the **smaller** when $\mathrm{off}_j = -1$.

Both L_j and off_j are explicit step functions of j, with jumps only at

$$1 < \ell_1 = \lceil \tfrac{n}{8} \rceil \le \beta_1 = \left\lceil \tfrac{n-2}{4} \right\rceil \le \ell_2 = \lceil \tfrac{n}{4} \rceil$$

$$\le \beta_2 = \left\lceil \tfrac{n+2}{4} \right\rceil < \ell_3 = \left\lceil \tfrac{3n}{8} \right\rceil < H < H+1,$$

giving **seven** intervals with $(L, \mathrm{off}) = (0,0), (1,-1), (1,0), (2,-1), (2,0), (3,-1), (3,0)$ in order. The last is the single point $j = H$, where $4H + 2 = 2n$ forces $q_H = 2$. Combining the two parities, an interval $[\alpha, \beta)$ has $\beta - \alpha$ tiles whose multiples total $\Phi(\beta) - \Phi(\alpha)$, independently of which parity starts it. Its contribution to the mixed count is $G - LC$ if $\mathrm{off} = 0$ and $(L+1)C - G$ if $\mathrm{off} = -1$, with $G = \Phi(\beta) - \Phi(\alpha)$ and $C = \beta - \alpha$.

Summing the seven intervals and telescoping the Φ's (note $\Phi(1) = 0$):

$$\mathrm{card}(\mathrm{mixed}) = 2\big[\Phi(\ell_1) - \Phi(\beta_1) + \Phi(\ell_2) - \Phi(\beta_2) + \Phi(\ell_3) - \Phi(H)\big] + \Phi(H+1) + \Sigma_C, \qquad (5.1)$$

$$\Sigma_C = 2(\beta_1 - \ell_1) - (\ell_2 - \beta_1) + 3(\beta_2 - \ell_2) - 2(\ell_3 - \beta_2) + 4(H - \ell_3) - 3.$$

Verification of (5.1). Zero failures against the direct count for every odd $9 \le n \le 20{,}001$.

5.5 Step 5: evaluating the seven Φ's

Two are immediate: $\Phi(H+1) = n - 1$ and $\Phi(H) = n - 5$. Three more are linear; writing $n = 8t + r$ with $r \in \{1, 3, 5, 7\}$,

$$-\Phi(\beta_1) + \Phi(\ell_2) - \Phi(\beta_2) = -2t + c_r, \qquad c_r = 3, -1, 1, -3.$$

The remaining two are the content, and they are handled together.

> **Lemma.** For odd $n = 8t + r \ge 51$, $\Phi(\lceil n/8 \rceil) + \Phi(\lceil 3n/8 \rceil) = \dfrac{5n - k_r}{8}$, $\qquad k_r = 13, 7, 25, 19$ for $r = 1, 3, 5, 7$.

Proof. Put $m = \lceil 3r/8 \rceil \in \{1, 2, 2, 3\}$, so $\lceil n/8 \rceil = t + 1$ and $\lceil 3n/8 \rceil = 3t + m$; the four arguments are $x \in \{t,\ t+1,\ 3t+m-1,\ 3t+m\}$. Because $8t \equiv -r \pmod n$,

$$8x \equiv e_x \pmod n, \qquad e_x = -r,\ 8 - r,\ 8m - 8 - 3r,\ 8m - 3r,$$

and **in every one of the four cases $\{|e_x|\}$ is a permutation of $\{1, 3, 5, 7\}$** — for instance $r = 1, m = 1$ gives $(-1, 7, -3, 5)$ and $r = 7, m = 3$ gives $(-7, 1, -5, 3)$. Since $32 \cdot 2x^2 = (8x)^2$,

$$32(2x^2 \bmod n) \equiv e_x^2 \pmod n, \qquad \sum_x e_x^2 = 1 + 9 + 25 + 49 = 84.$$

Write $32(2x^2 \bmod n) = e_x^2 + a_x n$. That a_x is an integer in $[1, 32)$ is the one place the hypothesis $n \ge 51$ is used: it gives $e_x^2 \le 49 < n$, so e_x^2 is already the reduced residue of the left side and $a_x \ge 0$, while $a_x = 0$ would force $32 \mid e_x^2$, false for $e_x^2 \in \{1, 9, 25, 49\}$; the upper bound is automatic from $2x^2 \bmod n < n$. So a_x is the unique such integer with $e_x^2 + a_x n \equiv 0 \pmod{32}$, i.e. $a_x \equiv -n^{-1} e_x^2 \pmod{32}$. Now $1^2, 3^2, 5^2, 7^2$ are $1, 9, 25, 17$ modulo 32, which are precisely the four classes there that are $\equiv 1 \pmod 8$; multiplying by the odd number $-n^{-1}$ sends $1 + 8k$ to $-n^{-1} + 8k'$ with k' running over $0, 1, 2, 3$ again, so the four values a_x are $c, c + 8, c + 16, c + 24$ for the single $c \in [1, 8]$

with $c \equiv -n^{-1} \pmod 8$. Since $n \equiv r$ and $r^2 \equiv 1 \pmod 8$, we have $c = 8 - r$, and therefore $\sum_x a_x = 4c + 48 = 80 - 4r$, whence

$$\sum_x (2x^2 \bmod n) = \frac{84 + (80 - 4r)n}{32} = \frac{(20 - r)n + 21}{8}.$$

On the other hand $\sum_x 2x^2 = 40t^2 + (24m - 8)t + (4m^2 - 4m + 4)$ exactly, and dividing the difference by $n = 8t + r$ gives $5t + s_r$ with $s_r = -1, 1, 0, 2$, which is $(5n - k_r)/8$. ■

Verification of the Lemma. Zero failures for every odd $n \le 300{,}001$ apart from $n = 3, 5, 7, 9, 17, 25, 49$. At each of those, $e_x^2 < n$ fails for one of the four x: that a_x comes out 32 below the value the congruence alone would give, the sum $\sum_x (2x^2 \bmod n)$ is short by exactly n, and the Lemma's stated value is 1 below the truth.

5.6 Step 6: assembly

Substituting $\ell_1 = t + 1$, $\beta_1 = 2t + \lceil \frac{r-2}{4} \rceil$, $\ell_2 = 2t + \lceil \frac{r}{4} \rceil$, $\beta_2 = 2t + \lceil \frac{r+2}{4} \rceil$, $\ell_3 = 3t + m$, $H = 4t + \frac{r-1}{2}$ into (5.1) together with the Lemma gives, in each of the four residue classes,

$$\text{card(mixed)} = 2t + 2 = 2A. \qquad ■$$

n	$8t+1$	$8t+3$	$8t+5$	$8t+7$
card(mixed)	$2t+2$	$2t+2$	$2t+2$	$2t+2$

Where the small values break, and why the two exception lists differ. Formula (5.1) itself holds for every odd $n \ge 9$, $n = 49$ included, so nothing goes wrong in Step 4 there; what fails below 51 is the evaluation in Step 5. At $n = 17$ and $n = 25$ the true value of $\Phi(\lceil n/8 \rceil) + \Phi(\lceil 3n/8 \rceil)$ exceeds the Lemma by 1 while the true linear part falls short of $-2t + c_r$ by 1, the two cancel in the bracket, and the conclusion survives. At $n = 49$ only the Lemma fails, the bracket is short by 1, and the count comes out $2A + 2 = 16$ against the predicted 14. At $n \le 9$ several of the evaluations fail at once. That is why the exception list of Theorem 5 and the exception list of the Lemma are not the same list.

5.7 Remark: what made it work

A local maximum is *two* consecutive conditions, while Theorem 2 counts j satisfying *one*; and while the intervals I_j tile, consecutive P_j overlap. The apparent obstruction dissolves in two moves. Steps 2 and 3 convert the two conditions into one — a condition on the count of multiples in the single interval P_j — and Step 4 observes that the P_j tile perfectly once separated by parity. The elementary weight $\sum_j \varepsilon_j = (n - 1)/2$, which follows from $\gcd(4, n) = 1$ because the increments $(4j + 2) \bmod n$ then run over every residue, is thereby joined by a second weight, of pairs rather than of single terms, and the same tiling carries both.

The one place where genuine arithmetic enters is the Lemma, and it enters through a small coincidence worth naming: the four numbers $8x \bmod n$ attached to $\lceil n/8 \rceil$ and $\lceil 3n/8 \rceil$ have signed representatives $e_x = \pm 1, \pm 3, \pm 5, \pm 7$ in some order, so $\sum_x e_x^2 = 84$ whatever n is. It is those representatives, not the reduced residues themselves, whose squares sum to 84.

6. Placement

Since $r_j = 2j^2 \bmod n$ is $n\{j^2\alpha\}$ with $\alpha = 2/n$ **rational**, the natural comparisons are these.

The linear analogue. For $\{j\alpha\}$ the three-distance theorem [1], proved independently by Sós, by Świerczkowski and by Surányi, states that the gaps between consecutive points take at most three values. Theorems 2 and 4 are of that flavour — a small fixed set of values with exactly determined multiplicities — but the object is quadratic and the statement is about the increments of the associated staircase rather than about gaps. We do not claim the three-distance theorem as a template; the resemblance is one of shape.

The quadratic literature runs the other way. For irrational α the fine-scale statistics of $\{j^2\alpha\}$ — pair correlation and spacings — are studied as limiting laws over a *continuum* of α. Heath-Brown [2] is quantitative as well as metric: he constructs a dense set of α for which the pair correlation is Poissonian, and for Diophantine α he bounds the pair correlation function by X with an error of size $X^{7/8}$ in the range $1 \le X \le \log N$ — a bound at a fixed scale rather than a limit. A statement of the present kind — a finite set of values with explicit multiplicities and no error term — uses the fact that $\alpha = 2/n$ is rational here, so that the orbit is finite and periodic. We use that structure; we do not claim it is the only route to such a statement.

The nearest relative is the literature on spacings of quadratic residues modulo q, for instance Kurlberg [3]. The results there are again distributional; Theorems 2, 4 and 5 are exact identities for a single explicit sequence.

We are not aware of the multiplicities of Theorem 2, the pair census of Theorem 4, or the count of Theorem 5 in the literature, and would be glad to be corrected.

Appendix A — Sample data

The increments W_j, $n = 7$: $0, 1, 1, 2, 3, 3, 4$. Here $A = \lfloor 14/8 \rfloor = 1$, and Theorem 2 predicts $\mathrm{card}\{0\} = \mathrm{card}\{4\} = 1$, $\mathrm{card}\{2\} = 2A - 1 = 1$, $\mathrm{card}\{1\} = \mathrm{card}\{3\} = (n+1)/2 - 2A = 2$ — exactly what the list shows. (Note that $n = 7$ is one of the five small exceptions to Theorem 5; the histogram of Theorem 2 has no exceptions at all.)

The residues $r_j = 2j^2 \bmod 17$: $0, 2, 8, 1, 15, 16, 4, 13, 9, 9, 13, 4, 16, 15, 1, 8, 2$. Interior local maxima at $j = 2, 5, 7, 10, 12, 15$ — six of them, and $2A = 2\lfloor 24/8 \rfloor = 6$.

The seven intervals of Step 4, $n = 101$ ($t = 12$, $r = 5$, $H = 50$): the boundaries are

$$1, \quad \ell_1 = 13, \quad \beta_1 = 25, \quad \ell_2 = 26, \quad \beta_2 = 26, \quad \ell_3 = 38, \quad H = 50, \quad H + 1 = 51,$$

so the fourth interval $[\ell_2, \beta_2)$ is empty here and six of the seven carry indices, with $(L, \mathrm{off}) = (0, 0)$ on $[1, 13)$, $(1, -1)$ on $[13, 25)$, $(1, 0)$ at $j = 25$, $(2, 0)$ on $[26, 38)$, $(3, -1)$ on $[38, 50)$ and $(3, 0)$ at $j = 50$.

Appendix B — Note on method

Two details were found by failure rather than by design and are recorded so that a reader reconstructing the argument does not lose time on them.

The constant 2 in the increment. In the application from which this paper was extracted, the increment appears as $H_j = 2 + W_j$ — the two extra strikes are the two full line-steps in the square

gap $4p + 8j + 4$, and the identity is proved in [P2, §0.1]. Dropping the additive constant breaks the formula in 57 of 78 tested cases (a count from an early run; no generator was kept). It is an artefact of that application, not of the staircase, but it is easy to mislay.

The seventh interval. A first version of (5.1) used six intervals and failed for essentially every n. The omission is the single point $j = H$, where $4H + 2 = 2n$ forces $q_H = 2$ rather than 1; with it restored the identity holds with zero failures on the whole tested range.

The computations and much of the prose in this paper were prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running the computations, and for auditing the papers against their own scripts. All statements were checked by the author, who is responsible for them; the repository README sets out the division of labour in full.

References

1. V. T. Sós, *On the distribution mod 1 of the sequence $n\alpha$*, Ann. Univ. Sci. Budapest, Eötvös Sect. Math. **1** (1958), 127–134.
2. D. R. Heath-Brown, *Pair correlation for fractional parts of αn^2*, Math. Proc. Cambridge Philos. Soc. **148** (2010), 385–407.
3. P. Kurlberg, *The distribution of spacings between quadratic residues, II*, Israel J. Math. **120** (2000), part A, 205–224.
4. MathOverflow, *Exact multiplicities for the increments of the staircase floor(2j^2/n) — known?*, question 514570 (2026), and the answer there giving the carry-count identity.